REMARKS ON FIVE QUESTIONS OF BAYART AND YAO ON COMPOSITION OPERATORS

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ABSTRACT. We answer five questions of Bayart and Yao on composition operators. On the Fréchet space $\mathcal{H}^\infty_+$ of uniformly convergent Dirichlet series, the spectrum of the composition operator induced by the vertical translation $\varphi(s) = s + i\tau$, $\tau \neq 0$, is the unit circle. On the Hardy space $\mathcal{H}^2$ of Dirichlet series, the symbol of a cyclic composition operator is univalent on the whole right half-plane. A diagonal operator with distinct eigenvalues λ_n has a minimal commutant if and only if no nonzero summable sequence (a_n) satisfies $\sum_n a_n \lambda_n^k = 0$ for every $k \geq 0$, and there are diagonal operators with distinct eigenvalues for which such a sequence exists. On the Hardy space of the unit disc, the assumption that ψ is zero-free can be removed from a theorem of Bayart and Yao on symbols $\varphi(z) = \lambda z^p \psi(z)$, and the composition operators induced by lens maps do not have a minimal commutant.

1. INTRODUCTION

For $\theta \in \mathbb{R}$ let $\mathbb{C}_\theta = \{s \in \mathbb{C} : \operatorname{Re} s > \theta\}$. Let $\mathcal{H}^2$ denote the Hilbert space of Dirichlet series $f(s) = \sum_{n\geq 1} a_n n^{-s}$ equipped with the norm

$$\|f\|_{\mathcal{H}^2} = \left(\sum_{n=1}^{\infty} |a_n|^2\right)^{1/2}.$$

This space was introduced by Hedenmalm, Lindqvist and Seip [8]. By the Cauchy–Schwarz inequality its elements are holomorphic in $\mathbb{C}_{1/2}$. Gordon and Hedenmalm [7] proved that a holomorphic map $\varphi\colon \mathbb{C}_{1/2} \to \mathbb{C}_{1/2}$ induces a bounded composition operator $C_\varphi f = f \circ \varphi$ on $\mathcal{H}^2$ if and only if φ belongs to the class $\mathcal{G}$ of maps $\varphi(s) = c_0 s + \psi(s)$, where c_0 is a nonnegative integer, ψ is a Dirichlet series which converges uniformly in $\mathbb{C}_\varepsilon$ for every $\varepsilon > 0$, and

(a) $\psi(\mathbb{C}_0) \subset \mathbb{C}_{1/2}$ if $c_0 = 0$;
(b) $\psi \equiv i\tau$ for some $\tau \in \mathbb{R}$, or $\psi(\mathbb{C}_0) \subset \mathbb{C}_0$, if $c_0 \geq 1$.

The integer c_0 is the *characteristic* of φ, and $\mathcal{G}_{>0}$ denotes the symbols of positive characteristic. We follow the notation of Bayart and Yao [2], whose

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results we cite with the numbering of arXiv:2406.19737v1, and we state their questions in the notation of the present paper.

Bayart and Yao constructed Königs maps for the symbols in $\mathcal{G}_{>0}$ and used them to study the spectrum, the cyclicity and the commutant of C_φ. For the spectrum, the natural analogue of the space of all holomorphic functions in the disc is the space $\mathcal{H}^\infty_+$ of Dirichlet series which converge uniformly in $\mathbb{C}_\varepsilon$ for every $\varepsilon > 0$, studied by Bonet [3]. It is a Fréchet space with the seminorms

$$P_\varepsilon(f) = \sup_{s\in\mathbb{C}_\varepsilon} |f(s)|, \qquad \varepsilon > 0.$$

Every $\varphi \in \mathcal{G}_{>0}$ maps each $\mathbb{C}_\varepsilon$ into itself, so C_φ is a continuous operator on $\mathcal{H}^\infty_+$. The spectrum $\sigma(C_\varphi)$ is the set of $\lambda \in \mathbb{C}$ such that $C_\varphi - \lambda I$ is not invertible on $\mathcal{H}^\infty_+$; by the open mapping theorem for Fréchet spaces, invertible means bijective. Bayart and Yao proved that C_φ is invertible on $\mathcal{H}^\infty_+$ if and only if $\varphi(s) = s + i\tau$ for some $\tau \in \mathbb{R}$ [2, Theorem 3.6], and they computed $\sigma(C_\varphi)$ for every other $\varphi \in \mathcal{G}_{>0}$ [2, Theorem 3.7]. For the vertical translations they found the point spectrum $\{m^{-i\tau} : m \in \mathbb{N}\}$ [2, Theorem 3.1(iii)] and asked the following.

Question (Bayart–Yao [2, Question 3.8])**.** *What is the spectrum of C_φ provided $\varphi(s) = s + i\tau$?*

They observed that this is already a delicate question for the composition operators induced by rotations on the space of holomorphic functions in the disc, which were studied by Bonet [4]. The difficulty is that the spectrum of an operator on a Fréchet space need not be closed, so the density of the point spectrum in the unit circle $\mathbb{T}$ does not settle the question. Our first result does.

Theorem 1. *Suppose that $\tau \in \mathbb{R} \setminus \{0\}$ and that $\varphi(s) = s + i\tau$. Then the spectrum of C_φ on $\mathcal{H}^\infty_+$ is $\mathbb{T}$. More precisely, for every $\lambda \in \mathbb{T}$ which is not of the form $m^{-i\tau}$ with $m \in \mathbb{N}$, the operator $C_\varphi - \lambda I$ is injective, and there is a Dirichlet series g which converges absolutely and is bounded on $\mathbb{C}_0$ such that g is not in the range of $C_\varphi - \lambda I$.*

For $\tau = 0$ the operator C_φ is the identity and its spectrum is $\{1\}$. Our second result concerns cyclicity on $\mathcal{H}^2$. Bayart and Yao proved that if $\varphi \in \mathcal{G}$ and C_φ is cyclic on $\mathcal{H}^2$, then $c_0 = 1$ [2, Proposition 4.1]. Their proof uses that point evaluations are bounded on $\mathcal{H}^2$ at the points of $\mathbb{C}_{1/2}$, and they asked whether it can be extended to the whole half-plane $\mathbb{C}_0$ on which φ is defined.

Question (Bayart–Yao [2, Question 4.2])**.** *The above proof shows that provided C_φ is cyclic, φ is univalent on $\mathbb{C}_{1/2}$. Is it also univalent on $\mathbb{C}_0$?*

The answer is yes, and in a quantitative form.

Theorem 2. *Suppose that $\varphi \in \mathcal{G}_{>0}$ is not univalent on $\mathbb{C}_0$. Then the orthogonal complement of the range of C_φ in $\mathcal{H}^2$ is infinite-dimensional.*

Corollary 1. *Suppose that $\varphi \in \mathcal{G}$ and that C_φ is cyclic on $\mathcal{H}^2$. Then $c_0 = 1$ and φ is univalent on $\mathbb{C}_0$.*

The remaining three questions concern the minimal commutant property. For a bounded operator A on a Hilbert space H, let $\{A\}'$ be the set of bounded operators which commute with A, let $\mathrm{alg}(A) = \{P(A) : P \in \mathbb{C}[X]\}$, and let $\overline{\mathrm{alg}(A)}^\sigma$ be the closure of $\mathrm{alg}(A)$ in the weak operator topology σ. Then $\overline{\mathrm{alg}(A)}^\sigma \subset \{A\}'$, and A has a *minimal commutant* if $\overline{\mathrm{alg}(A)}^\sigma = \{A\}'$. Lacruz, León-Saavedra, Petrovic and Rodríguez-Piazza [9] determined which composition operators with linear fractional symbols on the Hardy space $H^2(\mathbb{D})$ of the unit disc have a minimal commutant. Bayart and Yao proved that C_φ does not have a minimal commutant on $\mathcal{H}^2$ when $c_0 \geq 2$ and $\varphi(\mathbb{C}_0) \subset \mathbb{C}_\varepsilon$ for some $\varepsilon > 0$ [2, Theorem 1.2], and found a large class of symbols with $c_0 = 1$ for which it does [2, Theorem 7.1].

For a translation $\varphi(s) = s + c_1$ the operator C_φ is diagonal in the basis $(n^{-s})_{n\geq 1}$, with eigenvalues n^{-c_1}. This led Bayart and Yao to diagonal operators: if $(e_n)_{n\geq 1}$ is an orthonormal basis of H and (λ_n) is a bounded sequence of complex numbers, let D be the operator defined by $De_n = \lambda_n e_n$. If two of the λ_n coincide, then $\{D\}'$ is not commutative, so D does not have a minimal commutant. If (λ_n) is injective and either convergent or contained in a circle, then D has a minimal commutant [2, Proposition 5.4].

Question (Bayart–Yao [2, Question 5.6])**.** *Does there exist a diagonal operator associated to an injective sequence (λ_n) which fails to have a minimal commutant? If so, can we give a characterization of diagonal operators with a minimal commutant?*

The answer to the first question is yes, and the characterization is in terms of discrete measures which annihilate the polynomials.

Theorem 3. *Suppose that (λ_n) is bounded and injective. Then D has a minimal commutant if and only if the only sequence (a_n) in ℓ^1 such that*

$$\sum_{n=1}^{\infty} a_n \lambda_n^k = 0 \qquad \text{for every } k \geq 0 \tag{1}$$

is $a_n = 0$ for every n.

Theorem 4. *There is a bounded injective sequence (λ_n) in the unit disc such that D does not have a minimal commutant. More precisely, the orthogonal projection of H onto $\mathbb{C}e_1$ belongs to $\{D\}'$ but not to $\overline{\mathrm{alg}(D)}^\sigma$.*

The last two questions concern composition operators on $H^2(\mathbb{D})$. Write $\varphi^{[n]}$ for the nth iterate of a holomorphic self-map φ of $\mathbb{D}$. Bayart and Yao proved [2, Corollary 8.5] that C_φ does not have a minimal commutant on $H^2(\mathbb{D})$ if $\varphi(z) = \lambda z^p \psi(z)$ with $p \geq 2$, $\lambda \neq 0$, $\psi(0) = 1$, ψ does not vanish in $\mathbb{D}$, and $\|\varphi^{[n]}\|_\infty < 1$ for some $n \geq 1$. The zero-free assumption is used to solve Böttcher's equation $u \circ \varphi = \lambda u^p$ by a function u on $\mathbb{D}$ with $u'(0) \neq 0$ [2, Lemma 8.4].

Question (Bayart–Yao [2, Question 8.7])**.** *Can we dispense with the assumption that ψ does not vanish on $\mathbb{D}$ in Corollary 8.5?*

Theorem 5. *Suppose that $\varphi\colon \mathbb{D} \to \mathbb{D}$ is holomorphic and that $\varphi(z) = \lambda z^p \psi(z)$, where $p \geq 2$, $\lambda \neq 0$ and $\psi(0) = 1$. Suppose moreover that $\|\varphi^{[n]}\|_\infty < 1$ for some $n \geq 1$. Then there is a holomorphic self-map $\tilde{\varphi}$ of $\mathbb{D}$ which commutes with φ and satisfies $C_{\tilde{\varphi}} \notin \overline{\operatorname{alg}(C_\varphi)}^{\sigma}$. In particular, C_φ does not have a minimal commutant on $H^2(\mathbb{D})$.*

The final question concerns the lens maps. For $0 < \alpha < 1$ the lens map is

$$\varphi_\alpha(z) = \frac{\big((1+z)/(1-z)\big)^\alpha - 1}{\big((1+z)/(1-z)\big)^\alpha + 1}.$$

Its Königs function is $u(z) = \log\big((1+z)/(1-z)\big)$, which satisfies $u \circ \varphi_\alpha = \alpha u$ and maps $\mathbb{D}$ onto the strip

$$S = \{w \in \mathbb{C} : |\operatorname{Im} w| < \pi/2\}.$$

Lacruz, León-Saavedra, Petrovic and Rodríguez-Piazza [9, Theorem 6.1] (Theorem A in [2]) proved that if φ is a univalent self-map of $\mathbb{D}$ with $\varphi(0) = 0$ whose Königs function has a bounded range which is strictly starlike with respect to the origin, then C_φ has a minimal commutant on $H^2(\mathbb{D})$. The strip S is strictly starlike with respect to the origin, but it is unbounded.

Question (Bayart–Yao [2, Question 8.11])**.** *Does C_{φ_α} have a minimal commutant?*

The parameter range is printed as $0 < \alpha < \pi/2$ in [2]. Since $(1+z)/(1-z)$ maps $\mathbb{D}$ onto $\mathbb{C}_0$, the function $((1+z)/(1-z))^\alpha$ maps $\mathbb{D}$ onto the sector $|\arg w| < \alpha\pi/2$. Hence, for $0 < \alpha < \pi/2$, the formula defines a self-map of $\mathbb{D}$ exactly when $\alpha \leq 1$, and φ_1 is the identity. The usual range of the lens maps is $0 < \alpha < 1$, and we treat all of it.

Theorem 6. *Suppose that $0 < \alpha < 1$. Then C_{φ_α} does not have a minimal commutant on $H^2(\mathbb{D})$. More precisely, the operator $Rf(z) = f(-z)$ commutes with C_{φ_α}, but $R \notin \overline{\operatorname{alg}(C_{\varphi_\alpha})}^{\sigma}$.*

Theorem 6 shows that the boundedness of the range cannot be dropped from the theorem of Lacruz, León-Saavedra, Petrovic and Rodríguez-Piazza, even when C_φ is compact and every power of the Königs function belongs to $H^2(\mathbb{D})$; see Section 5. To the best of our knowledge, Theorems 1, 2, 5 and 6 are new. Theorem 3 is the diagonal case of Sarason's description of the weakly closed algebra generated by a normal operator, and the example behind Theorem 4 goes back to Wolff; we include short self-contained proofs and discuss the history in Section 5.

Let us briefly describe our approach. On coefficients, $C_\varphi - \lambda I$ with $\varphi(s) = s + i\tau$ is multiplication by $n^{-i\tau} - \lambda$. For $\lambda \in \mathbb{T}$ these multipliers come arbitrarily close to 0, so solving $(C_\varphi - \lambda I)f = g$ forces large coefficients in f, while the coefficients of an element of $\mathcal{H}^\infty_+$ grow more slowly than every

power n^σ with $\sigma > 0$. This is Theorem 1. For Theorem 2 we follow the proof of [2, Proposition 4.1], where a collision $\varphi(a) = \varphi(b)$ in $\mathbb{C}_{1/2}$ gives the vector $K_a - K_b$ in the kernel of C_φ^*. Point evaluations are unbounded on $\mathcal{H}^2$ at the points of $\mathbb{C}_0 \setminus \mathbb{C}_{1/2}$, so we replace them by evaluations of the part of f supported on the integers generated by the first N primes. These functionals are bounded on $\mathcal{H}^2$ at every point of $\mathbb{C}_0$, and they intertwine C_φ with a composition operator whose symbol φ_N tends to φ as $N \to \infty$; by Hurwitz's theorem φ_N inherits the collision of φ. The proofs of Theorems 3 and 6 exhibit a functional $T \mapsto \langle Tx, y\rangle$ which vanishes on every power of the operator but not on a given element of the commutant. For diagonal operators such functionals are exactly the discrete measures $\sum_n a_n \delta_{\lambda_n}$ with $(a_n) \in \ell^1$, and the condition is that the measure annihilates the polynomials. For the lens maps the commuting operator is the reflection $Rf(z) = f(-z)$, x is the bounded function $1/(u^3 - 64i)$, and y comes from the Cauchy transform of a function on the positive half-line which vanishes at the points $64i\alpha^{-3k}$, $k \geq 0$, but not at $-64i$. For Theorem 5 we use the univalent Böttcher coordinate only in a small disc, and compose it with an iterate of φ to obtain a global solution of Böttcher's equation; in contrast to [2], this global solution is not univalent near the origin, and we compensate for this by a variant of [2, Lemma 8.1].

Organization. The present paper is organized into four additional sections. In Section 2 we collect two facts used throughout. In Section 3 we prove Theorems 1 and 2 on composition operators on spaces of Dirichlet series. In Section 4 we prove Theorems 3 to 6 on minimal commutants. Section 5 contains concluding remarks.

2. Preliminaries

Coefficients of uniformly convergent Dirichlet series. By a theorem of Bohr, a Dirichlet series which converges in $\mathbb{C}_0$ and is bounded in $\mathbb{C}_\varepsilon$ for every $\varepsilon > 0$ converges uniformly in each $\mathbb{C}_\varepsilon$; see [2, Section 2.1] and [3]. We will use the following standard bound for the coefficients.

Lemma 1. *Suppose that $f(s) = \sum_{n\geq 1} a_n n^{-s}$ is in $\mathcal{H}^\infty_+$ and that $\sigma > 0$. Then $|a_n| \leq n^\sigma P_{\sigma/2}(f)$ for every $n \geq 1$.*

Proof. Since the series converges uniformly on the line $\operatorname{Re} s = \sigma$, we may integrate it term by term against n^{it}. This gives

$$a_n n^{-\sigma} = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} f(\sigma + it) n^{it} \, \mathrm{d}t.$$

The modulus of the right-hand side is at most $\sup_{t\in\mathbb{R}} |f(\sigma+it)| \leq P_{\sigma/2}(f)$. □

The weak operator topology. For vectors x, y in a Hilbert space H, the functional $T \mapsto \langle Tx, y\rangle$ on the bounded operators on H is continuous in the weak operator topology. Hence, if $\Lambda(T) = \langle Tx, y\rangle$ vanishes on A^k for every

$k \geq 0$, then Λ vanishes on $\overline{\mathrm{alg}(A)}^{\sigma}$, and an operator B with $\Lambda(B) \neq 0$ does not belong to $\overline{\mathrm{alg}(A)}^{\sigma}$. We will use this in the proofs of Theorems 3 and 6.

3. Composition operators on spaces of Dirichlet series

3.1. **The spectrum of vertical translations.** Throughout this subsection $\tau \neq 0$, $\varphi(s) = s + i\tau$ and $C = C_\varphi$. In terms of coefficients,

$$Cf(s) = \sum_{n=1}^{\infty} a_n n^{-i\tau} n^{-s}.$$

Comparing coefficients shows that the eigenvalues of C are the numbers $m^{-i\tau}$, $m \in \mathbb{N}$, with eigenvectors m^{-s}; this is [2, Theorem 3.1(iii)].

The proof of Theorem 1 rests on the following observation. The operator $C - \lambda I$ multiplies the nth coefficient by $n^{-i\tau} - \lambda = e^{-i\tau \log n} - \lambda$. Since consecutive values of $\log n$ differ by about $1/n$, the points $n^{-i\tau}$ wind around $\mathbb{T}$ ever more slowly, and for $\lambda \in \mathbb{T}$ there are frequencies n with $|n^{-i\tau} - \lambda| \lesssim 1/n$. Dividing by these small multipliers, a preimage of a bounded Dirichlet series supported on such frequencies would need coefficients of size about n. This contradicts Lemma 1, which allows only coefficients of size n^σ for every $\sigma > 0$. The same mechanism does not work for $|\lambda| \neq 1$, since then $|n^{-i\tau} - \lambda| \geq |1 - |\lambda||$.

Proof of Theorem 1. We first show that $\sigma(C) \subset \mathbb{T}$. Since $s \mapsto s + i\tau$ maps each $\mathbb{C}_\varepsilon$ onto itself, $P_\varepsilon(Cf) = P_\varepsilon(f)$ for every $f \in \mathcal{H}^\infty_+$ and every $\varepsilon > 0$. Moreover C is invertible, and its inverse $C_{s-i\tau}$ has the same property. Suppose that $|\lambda| > 1$. Then the series $-\sum_{n\geq 0} \lambda^{-n-1} C^n$ converges in every seminorm, since $P_\varepsilon(\lambda^{-n-1} C^n f) = |\lambda|^{-n-1} P_\varepsilon(f)$. As $\mathcal{H}^\infty_+$ is complete, its sum is a continuous operator, and it is the inverse of $C - \lambda I$. If $0 < |\lambda| < 1$, we write $C - \lambda I = C(I - \lambda C^{-1})$ and invert $I - \lambda C^{-1}$ by the series $\sum_{n\geq 0} \lambda^n C^{-n}$ in the same way. Finally $0 \notin \sigma(C)$, since C is invertible.

The eigenvalues $m^{-i\tau}$ lie in $\sigma(C)$. Let now $\lambda = e^{i\theta}$ be a point of $\mathbb{T}$ which is not an eigenvalue. Then $C - \lambda I$ is injective, and we will show that it is not onto. For an integer m put $x_m = \exp\big((2\pi m - \theta)/\tau\big)$, so that $x_m^{-i\tau} = e^{i\theta} = \lambda$. Let $m_1, m_2, \dots$ be the integers m with $x_m \geq e$, listed so that x_{m_k} increases. Then $x_{m_k} = x_{m_1} \rho^{k-1}$ with $\rho = e^{2\pi/|\tau|} > 1$. Put $n_k = \lfloor x_{m_k} \rfloor$. After discarding finitely many k, the integers n_k are distinct and $n_k \geq c\rho^k$ for some $c > 0$. Since $0 \leq \log x_{m_k} - \log n_k \leq \log(1 + 1/n_k) \leq 1/n_k$, we find that

$$|n_k^{-i\tau} - \lambda| = \big|e^{-i\tau \log n_k} - e^{-i\tau \log x_{m_k}}\big| \leq \frac{|\tau|}{n_k}. \tag{2}$$

Let

$$g(s) = \sum_{k\geq 1} k^{-2} n_k^{-s}.$$

This series converges absolutely and uniformly in $\mathbb{C}_0$, and $|g| \leq \pi^2/6$ there; in particular $g \in \mathcal{H}^\infty_+$.

Suppose that $f(s) = \sum_n a_n n^{-s}$ is in $\mathcal{H}^\infty_+$ and that $(C - \lambda I)f = g$. Comparing the coefficients of n_k^{-s}, we get $(n_k^{-i\tau} - \lambda)a_{n_k} = k^{-2}$. By (2), $|a_{n_k}| \geq n_k/(|\tau|k^2)$. On the other hand, Lemma 1 with $\sigma = 1/2$ gives $|a_{n_k}| \leq n_k^{1/2} P_{1/4}(f)$. Hence

$$P_{1/4}(f) \geq \frac{n_k^{1/2}}{|\tau|k^2} \geq \frac{c^{1/2}\rho^{k/2}}{|\tau|k^2}$$

for every k, which is absurd. Thus g is not in the range of $C - \lambda I$, and $\lambda \in \sigma(C)$. □

3.2. **Cyclicity and univalence on the half-plane.** Let $p_1 < p_2 < \cdots$ be the primes, fix $N \geq 1$, and let S_N be the set of positive integers all of whose prime factors lie in $\{p_1, \ldots, p_N\}$. For a Dirichlet series $f(s) = \sum_n a_n n^{-s}$ put

$$A_N f(s) = \sum_{n \in S_N} a_n n^{-s}.$$

Since $mn \in S_N$ if and only if both m and n are in S_N, the map A_N is multiplicative on formal Dirichlet series: $A_N(fg) = A_N f \cdot A_N g$.

For $f \in \mathcal{H}^2$ and $\sigma > 0$, the Cauchy–Schwarz inequality and the Euler product give

$$\sum_{n \in S_N} |a_n| n^{-\sigma} \leq \|f\|_{\mathcal{H}^2} \prod_{j=1}^{N} (1 - p_j^{-2\sigma})^{-1/2}. \tag{3}$$

Hence $A_N f$ converges absolutely in $\mathbb{C}_0$, and for every $s \in \mathbb{C}_0$ the functional $\ell_s(f) = A_N f(s)$ is bounded on $\mathcal{H}^2$. Write $z(s) = (p_1^{-s}, \ldots, p_N^{-s})$, a point of the polydisc $\mathbb{D}^N$. If $n = p_1^{\kappa_1} \cdots p_N^{\kappa_N}$, then $\ell_s(n^{-s}) = z(s)^\kappa$, while $\ell_s(n^{-s}) = 0$ for $n \notin S_N$. Thus ℓ_s depends only on $z(s)$. In terms of reproducing kernels, $\ell_s(f) = \langle f, K_s^N \rangle$, where

$$K_s^N(w) = \sum_{n \in S_N} n^{-\overline{s}} n^{-w} = \prod_{j=1}^{N} (1 - p_j^{-\overline{s}-w})^{-1}$$

is the reproducing kernel $K_s(w) = \zeta(\overline{s} + w)$ of $\mathcal{H}^2$ with the primes p_j, $j > N$, removed from the Euler product. The kernel K_s belongs to $\mathcal{H}^2$ only for $s \in \mathbb{C}_{1/2}$, while K_s^N belongs to $\mathcal{H}^2$ for every $s \in \mathbb{C}_0$. In the proof of [2, Proposition 4.1], a collision $\varphi(a) = \varphi(b)$ with $a, b \in \mathbb{C}_{1/2}$ gives $C_\varphi^*(K_a - K_b) = K_{\varphi(a)} - K_{\varphi(b)} = 0$. We will show that a collision $\varphi_N(a) = \varphi_N(b)$ of an approximating symbol φ_N with $a, b \in \mathbb{C}_0$ gives $C_\varphi^*(K_a^N - K_b^N) = 0$. The next lemma is the reason for working with A_N: it does not increase the supremum of a Dirichlet polynomial on vertical lines.

Lemma 2. *Suppose that P is a Dirichlet polynomial and that $\sigma > 0$. Then*

$$\sup_{t \in \mathbb{R}} |A_N P(\sigma + it)| \leq \sup_{t \in \mathbb{R}} |P(\sigma + it)|.$$

Proof. Let p_K be the largest prime which divides one of the frequencies of P, where we may assume that $K \geq N$, and write $P(s) = Q(p_1^{-s}, \dots, p_K^{-s})$ for a polynomial Q in K variables. Integrating over the variables $w_{N+1}, \dots, w_K$ in $\mathbb{T}$ removes every monomial which contains one of them, so

$$A_N P(\sigma + it) = \int_{\mathbb{T}^{K-N}} Q(p_1^{-\sigma-it}, \dots, p_N^{-\sigma-it}, p_{N+1}^{-\sigma} w_{N+1}, \dots, p_K^{-\sigma} w_K)\, \mathrm{d}m_{K-N}(w),$$

where m_{K-N} is the normalized Haar measure. Since $\log p_1, \dots, \log p_K$ are linearly independent over $\mathbb{Q}$, Kronecker's theorem [12, Theorem 2.2.4] shows that the points $(p_1^{-it}, \dots, p_K^{-it})$, $t \in \mathbb{R}$, are dense in $\mathbb{T}^K$. By continuity, the modulus of the integrand is at most $\sup_t |P(\sigma + it)|$ at every point. □

Let now $\varphi(s) = c_0 s + \psi(s)$ be in $\mathcal{G}_{>0}$, write $\psi(s) = \sum_{n \geq 1} c_n n^{-s}$, and let $P_M(s) = \sum_{n \leq M} c_n n^{-s}$. Fix $\varepsilon > 0$. The polynomials P_M converge to ψ uniformly in $\mathbb{C}_\varepsilon$, so $B_\varepsilon = \sup_M \sup_{\mathbb{C}_\varepsilon} |P_M|$ is finite. The partial sums of $A_N \psi$ are the polynomials $A_N P_M$, and Lemma 2 applied to $P_M - P_{M'}$ shows that they are uniformly Cauchy in $\mathbb{C}_\varepsilon$. Hence $\psi_N = A_N \psi$ converges uniformly in each $\mathbb{C}_\varepsilon$, it is holomorphic in $\mathbb{C}_0$, it depends on s only through $z(s)$, and $\sup_{\mathbb{C}_\varepsilon} |\psi_N| \leq B_\varepsilon$ for every N. Moreover $|c_n| \leq Cn$, because the series of ψ converges at $s = 1$. Hence $|\psi_N(s) - \psi(s)| \leq \sum_{n > p_N} |c_n| n^{-3} \to 0$ for $\operatorname{Re} s \geq 3$. By Vitali's theorem, applied in each $\mathbb{C}_\varepsilon$,

$$\psi_N \to \psi \qquad \text{locally uniformly in } \mathbb{C}_0 \tag{4}$$

as $N \to \infty$. Put $\varphi_N(s) = c_0 s + \psi_N(s)$. Then $\varphi_N \to \varphi$ locally uniformly in $\mathbb{C}_0$, and φ_N is not constant, since $c_0 \geq 1$ and ψ_N is bounded in $\mathbb{C}_1$. The following lemma says that the functionals ℓ_s intertwine C_φ with composition by φ_N.

Lemma 3. *Suppose that $f(s) = \sum_{n \leq L} a_n n^{-s}$ is a Dirichlet polynomial and that $s \in \mathbb{C}_0$. Then*

$$\ell_s(C_\varphi f) = \sum_{n \in S_N,\, n \leq L} a_n n^{-\varphi_N(s)}.$$

In particular, if $s, t \in \mathbb{C}_0$ and $\varphi_N(s) = \varphi_N(t)$, then $\ell_s - \ell_t$ vanishes on the range of C_φ.

Proof. Write $\psi = c_1 + \psi^0$. The series of ψ^0 converges absolutely in $\mathbb{C}_2$, so for $\mu > 0$ the expansion

$$e^{-\mu\psi} = e^{-\mu c_1} \sum_{k=0}^{\infty} \frac{(-\mu\psi^0)^k}{k!},$$

multiplied out, is an absolutely convergent Dirichlet series in $\mathbb{C}_2$. Hence $f \circ \varphi = \sum_n a_n n^{-c_0 s} e^{-(\log n)\psi}$ is given in $\mathbb{C}_2$ by an absolutely convergent Dirichlet series, obtained by formal multiplication. By the uniqueness of the coefficients of a Dirichlet series, this series is $C_\varphi f$. We apply the multiplicative map A_N term by term. The frequencies of the term $n^{-c_0 s} e^{-(\log n)\psi}$

are the integers $n^{c_0}m$, where m runs through the frequencies of $e^{-(\log n)\psi}$. Since $c_0 \geq 1$, such an integer lies in S_N if and only if both n and m do. Hence a term with $n \notin S_N$ contributes nothing, while for $n \in S_N$ the term becomes $n^{-c_0 s} A_N(e^{-(\log n)\psi}) = n^{-c_0 s} e^{-(\log n)\psi_N}$; here A_N commutes with the exponential series because each coefficient of $e^{-\mu\psi}$ involves only finitely many powers of ψ^0. Thus $A_N(C_\varphi f)(s) = \sum_{n \in S_N} a_n n^{-\varphi_N(s)}$ for $s \in \mathbb{C}_2$. Both sides are holomorphic in $\mathbb{C}_0$, the left-hand side by (3), so they agree in $\mathbb{C}_0$.

If $\varphi_N(s) = \varphi_N(t)$, then $\ell_s - \ell_t$ vanishes on $C_\varphi f$ for every Dirichlet polynomial f. These vectors are dense in the range of C_φ, since C_φ is bounded, and $\ell_s - \ell_t$ is bounded. □

We are now ready to prove Theorem 2. We first move a collision of φ to a collision of φ_N by Hurwitz's theorem, and then use the open mapping theorem, as in [2, Proposition 4.1], to obtain infinitely many collisions of φ_N and hence infinitely many independent vectors $K^N_{a_k} - K^N_{b_k}$ orthogonal to the range of C_φ.

Proof of Theorem 2. Let $a \neq b$ be points of $\mathbb{C}_0$ with $\varphi(a) = \varphi(b)$. The function $\varphi - \varphi(b)$ is not constant and vanishes at a. Choose a closed disc $\Delta \subset \mathbb{C}_0$ centred at a which does not contain b, such that $\varphi - \varphi(b)$ does not vanish on $\partial\Delta$. By (4), $\varphi_N - \varphi_N(b) \to \varphi - \varphi(b)$ uniformly on Δ, so Hurwitz's theorem [5, Theorem VII.2.5] gives an N and a point $a' \in \Delta$ with $\varphi_N(a') = \varphi_N(b)$. Fix this N. Note that $a' \neq b$.

We claim that $z(a') \neq z(b)$. Otherwise $\psi_N(a') = \psi_N(b)$, since ψ_N depends only on z, and then $c_0 a' = c_0 b$, so $a' = b$.

Choose disjoint discs $U \ni a'$ and $V \ni b$ in $\mathbb{C}_0$, of radius less than $\pi/\log 2$, so small that $z(U) \cap z(V)$ is empty. Then z is injective on U and on V, since already $s \mapsto 2^{-s}$ is. The set $\varphi_N(V)$ is an open neighbourhood of $\varphi_N(b) = \varphi_N(a')$, and φ_N takes each value only finitely many times near a'. By the continuity of φ_N at a', there are points $a_k \in U$, $k \geq 1$, such that the values $\varphi_N(a_k)$ are pairwise distinct and lie in $\varphi_N(V)$. Choose $b_k \in V$ with $\varphi_N(b_k) = \varphi_N(a_k)$. The points $a_1, b_1, a_2, b_2, \ldots$ are pairwise distinct, and hence so are their images under z.

By Lemma 3, each $\Lambda_k = \ell_{a_k} - \ell_{b_k}$ is a bounded functional on $\mathcal{H}^2$ which vanishes on the range of C_φ. On the Dirichlet polynomials of the form $f(s) = F(p_1^{-s}, \ldots, p_N^{-s})$, where F is a polynomial, we have $\ell_s(f) = F(z(s))$. Since a polynomial can take arbitrary values at finitely many distinct points of $\mathbb{C}^N$, the functionals Λ_k are linearly independent. Hence the orthogonal complement of the range of C_φ is infinite-dimensional. □

Proof of Corollary 1. The closure of the range of a cyclic operator has codimension at most one, as noted in the proof of [2, Proposition 4.1]. By that proposition $c_0 = 1$, and by Theorem 2 the symbol φ is univalent on $\mathbb{C}_0$. □

4. Minimal commutants

4.1. **Diagonal operators.** Let (e_n), (λ_n) and D be as in the introduction, with (λ_n) bounded and injective. Since $\ker(D-\lambda_n)=\mathbb{C}e_n$, every $T\in\{D\}'$ satisfies $Te_n=\mu_n e_n$ for some μ_n with $|\mu_n|\le\|T\|$. Conversely every bounded diagonal operator commutes with D. Hence $\{D\}'$ consists of the operators $D_\mu e_n=\mu_n e_n$ with $\mu\in\ell^\infty$. For $m\ge 1$, let E_m be the orthogonal projection onto $\mathbb{C}e_m$; it belongs to $\{D\}'$.

Let us explain the idea behind Theorem 3. For $x,y\in H$ we have $\langle D_\mu x,y\rangle=\sum_n\mu_n a_n$ with $a_n=x_n\overline{y_n}$, and every $(a_n)\in\ell^1$ arises in this way. Thus the functionals which define the weak operator topology on $\{D\}'$ are the pairings of ℓ^∞ with ℓ^1, that is, integration against the discrete measures $\sum_n a_n\delta_{\lambda_n}$, and $\overline{\mathrm{alg}(D)}^\sigma$ corresponds to the weak-star closure in ℓ^∞ of the sequences $(P(\lambda_n))_{n\ge1}$. By duality this closure is all of ℓ^∞ unless a nonzero such measure annihilates the polynomials. Bayart and Yao [2, Theorem 5.3] gave a sufficient condition in terms of approximate interpolation by polynomials which are uniformly bounded on $\{\lambda_n : n\ge 1\}$; Theorem 3 replaces it by the dual condition, which is both necessary and sufficient.

Proof of Theorem 3. Suppose first that $(a_n)\in\ell^1$ is not identically zero and satisfies (1). Put $x_n=|a_n|^{1/2}$, and put $y_n=\overline{a_n}|a_n|^{-1/2}$ if $a_n\ne0$ and $y_n=0$ otherwise. Then $x=\sum_n x_n e_n$ and $y=\sum_n y_n e_n$ are in H, and $x_n\overline{y_n}=a_n$. Let $\Lambda(T)=\langle Tx,y\rangle$. For every $k\ge0$,

$$\Lambda(D^k)=\sum_{n=1}^\infty\lambda_n^k x_n\overline{y_n}=\sum_{n=1}^\infty a_n\lambda_n^k=0.$$

Choose m with $a_m\ne0$. Then $\Lambda(E_m)=a_m\ne0$, so $E_m\notin\overline{\mathrm{alg}(D)}^\sigma$, and D does not have a minimal commutant.

Suppose now that (1) has no nonzero solution in ℓ^1. Let V be the subspace of ℓ^∞ consisting of the sequences $(P(\lambda_n))_{n\ge1}$, $P\in\mathbb{C}[X]$. An element of ℓ^1 which annihilates V solves (1), so it is zero. By the Hahn–Banach theorem for the weak-star topology $\sigma(\ell^\infty,\ell^1)$, the subspace V is weak-star dense in ℓ^∞. Let $D_\mu\in\{D\}'$, and choose a net of polynomials P_β such that $(P_\beta(\lambda_n))_{n\ge1}\to\mu$ in the weak-star topology. For $x,y\in H$ the sequence $(x_n\overline{y_n})$ is in ℓ^1, so

$$\langle P_\beta(D)x,y\rangle=\sum_{n=1}^\infty P_\beta(\lambda_n)x_n\overline{y_n}\to\sum_{n=1}^\infty\mu_n x_n\overline{y_n}=\langle D_\mu x,y\rangle.$$

Thus $P_\beta(D)\to D_\mu$ in the weak operator topology. Hence $\{D\}'\subset\overline{\mathrm{alg}(D)}^\sigma$, and the reverse inclusion always holds. □

For $|z|>\sup_n|\lambda_n|$ we have $\sum_n a_n/(z-\lambda_n)=\sum_{k\ge0}z^{-k-1}\sum_n a_n\lambda_n^k$. Hence (1) holds if and only if the Cauchy transform $\sum_n a_n/(z-\lambda_n)$ vanishes in a neighbourhood of infinity. The next example of such a sequence goes back to Wolff [16]. The idea is that normalized area measure on $\mathbb{D}$ has the same moments $\int w^k$ as δ_0, by the mean value property. If we cut $\mathbb{D}$ into small

discs, up to a null set, and replace the area measure of each disc by a point mass at its centre, the mean value property shows that the moments do not change. The resulting discrete measure minus δ_0 annihilates the polynomials.

Proof of Theorem 4. By the Vitali covering theorem [6, Section 1.5, Corollary 1], the unit disc contains countably many pairwise disjoint closed discs $B_j = \overline{D(c_j, r_j)}$, $j \geq 1$, whose union has full area measure in $\mathbb{D}$. Let $\mathrm{d}A$ denote area measure. The mean value property of w^k on each B_j gives

$$\sum_{j=1}^{\infty} r_j^2 c_j^k = \frac{1}{\pi} \sum_{j=1}^{\infty} \int_{B_j} w^k \, \mathrm{d}A(w) = \frac{1}{\pi} \int_{\mathbb{D}} w^k \, \mathrm{d}A(w) = \begin{cases} 1, & k = 0, \\ 0, & k \geq 1, \end{cases}$$

where the series converge absolutely since $|c_j| < 1$ and $\sum_j r_j^2 = 1$. Hence the measure $\nu = \sum_j r_j^2 \delta_{c_j} - \delta_0$ annihilates every polynomial. The centres c_j are distinct, since the discs are disjoint, and infinitely many of them differ from 0, since each $r_j < 1$. Thus $\nu \neq 0$; if one c_j equals 0, its mass merges with $-\delta_0$ into $r_j^2 - 1 \neq 0$. Enumerate the support of ν as $\lambda_1 = 0, \lambda_2, \lambda_3, \ldots$ and let $a_n = \nu(\{\lambda_n\})$. Then (λ_n) is injective and bounded by 1, the sequence (a_n) is in ℓ^1, every a_n is nonzero, and (1) holds. The first part of the proof of Theorem 3, with $m = 1$, shows that $E_1 \notin \overline{\mathrm{alg}(D)}^{\sigma}$. □

Theorem 3 also recovers the positive results of [2, Proposition 5.4].

Corollary 2. *Suppose that (λ_n) is bounded and injective, and let K be the closure of $\{\lambda_n : n \geq 1\}$. If K has empty interior and connected complement, or if every λ_n lies on a fixed circle, then D has a minimal commutant.*

Proof. In the first case, Mergelyan's theorem [13, Theorem 20.5] says that the polynomials are dense in $C(K)$, so a measure on K which annihilates the polynomials is zero. In the second case we may assume, after an affine change of variable, that the circle is $\mathbb{T}$. By the F. and M. Riesz theorem [13, Theorem 17.13], a measure ν on $\mathbb{T}$ with $\int z^k \, \mathrm{d}\nu = 0$ for every $k \geq 1$ is absolutely continuous with respect to arc length, and a discrete such measure is zero. In both cases Theorem 3 applies. □

A convergent sequence has a countable closure, which has empty interior and connected complement, so Corollary 2 contains [2, Proposition 5.4, parts (ii) and (iii)].

4.2. **Symbols with a superattracting fixed point.** In this subsection φ is as in Theorem 5. Since $\psi(0) = 1$, the map φ vanishes at the origin to order exactly p. The commuting map that we construct vanishes at the origin to an order which lies in $p\mathbb{N}$, so [2, Lemma 8.1], which requires the order to be outside $p\mathbb{N}$, does not apply. We first record the variant that we need.

Lemma 4. *Suppose that $\tilde{\varphi}$ is a holomorphic self-map of $\mathbb{D}$ which vanishes at the origin to an order q which is not a power of p. Then $C_{\tilde{\varphi}} \notin \overline{\mathrm{alg}(C_\varphi)}^{\sigma}$.*

Proof. The iterate $\varphi^{[j]}$ vanishes at the origin to order exactly p^j. Let $d_{j,m} = \langle \varphi^{[j]}, z^m \rangle$, so that $d_{j,m} = 0$ for $m < p^j$ and $d_{j,p^j} \neq 0$. Since q is not a power of p, we have $q \geq 2$, and there is an $l \geq 0$ with $p^l < q < p^{l+1}$.

Suppose that a net of polynomials $P_\beta(X) = \sum_j a_{\beta,j} X^j$ satisfies $P_\beta(C_\varphi) \to C_{\tilde{\varphi}}$ in the weak operator topology. Applying the operators to the function z and pairing with z^m we get

$$\sum_j a_{\beta,j} d_{j,m} = \langle P_\beta(C_\varphi) z, z^m \rangle \to \langle \tilde{\varphi}, z^m \rangle.$$

Take $m = p^i$ with $0 \leq i \leq l$. Then only the indices $j \leq i$ contribute, and $\langle \tilde{\varphi}, z^{p^i} \rangle = 0$, since $p^i < q$. As $d_{i,p^i} \neq 0$, induction on i gives $a_{\beta,i} \to 0$ for $i = 0, 1, \ldots, l$. Now take $m = q$. Only the indices j with $p^j \leq q$, that is $j \leq l$, contribute, so the left-hand side tends to 0. This contradicts $\langle \tilde{\varphi}, z^q \rangle \neq 0$. □

If $q \geq 2$ is not in $p\mathbb{N}$, then q is not a power of p, so Lemma 4 contains [2, Lemma 8.1] in the case where $\tilde{\varphi}(0) = 0$.

Let us compare with the approach of Bayart and Yao. In [2, Theorem 8.2] the commuting map is $\tilde{\varphi} = u^{-1} \circ (\mu u^q)$, where u solves Böttcher's equation on all of $\mathbb{D}$ and is univalent near the origin, and the zero-free assumption is used to construct such a u. We split these two roles between two functions: the local Böttcher coordinate B, which is univalent and is the only function we invert, and the global solution $v = B \circ \varphi^{[k]}$, which need not be univalent. The price is that $\tilde{\varphi}$ vanishes to an order in $p\mathbb{N}$, which is why we need Lemma 4.

Proof of Theorem 5. By Böttcher's theorem [11, Theorem 9.1], there are $r \in (0,1)$ and a univalent holomorphic function B on the disc $D(0,r)$ with $B(0) = 0$ and $B \circ \varphi = B^p$ near the origin. (In the normalization of [2], the function $u = cB$ with $c^{p-1} = 1/\lambda$ satisfies $u \circ \varphi = \lambda u^p$.) By the Schwarz lemma $|\varphi(z)| \leq |z|$, so φ maps $D(0,r)$ into itself, and by the identity theorem

$$B \circ \varphi = B^p \qquad \text{in } D(0,r). \tag{5}$$

Fix $\delta \in (0,1)$ such that $D(0,\delta) \subset B(D(0,r))$, and let $B^{-1} \colon B(D(0,r)) \to D(0,r)$ be the inverse map.

We next show that $\|\varphi^{[k]}\|_\infty \to 0$ as $k \to \infty$. Let $c = \|\varphi^{[n]}\|_\infty < 1$. The function $\varphi(z)/z$ is holomorphic in $\mathbb{D}$, bounded by 1, and vanishes at the origin, so by the maximum principle $\gamma = \max_{|z| \leq c} |\varphi(z)/z| < 1$. The closed disc $\overline{D(0,c)}$ is mapped into itself by φ, so $|\varphi^{[n+j]}(z)| \leq \gamma^j c$ for every $z \in \mathbb{D}$ and every $j \geq 0$.

Since B is continuous at the origin, we can therefore fix $k \geq 1$ such that $\varphi^{[k]}(\mathbb{D}) \subset D(0,r)$ and such that the function $v = B \circ \varphi^{[k]}$ satisfies $\sup_{\mathbb{D}} |v| < \delta$. For $z \in \mathbb{D}$ the point $\varphi^{[k]}(z)$ is in $D(0,r)$, so (5) gives

$$v(\varphi(z)) = B\big(\varphi(\varphi^{[k]}(z))\big) = B\big(\varphi^{[k]}(z)\big)^p = v(z)^p. \tag{6}$$

Since $B'(0) \neq 0$, the function v vanishes at the origin to order exactly p^k. Put

$$\tilde{\varphi} = B^{-1} \circ v^{p+1}.$$

This is a holomorphic map from $\mathbb{D}$ into $D(0,r)$, since $|v^{p+1}| \le |v| < \delta$. For $z \in \mathbb{D}$, both $\tilde\varphi(\varphi(z))$ and $\varphi(\tilde\varphi(z))$ lie in $D(0,r)$, and by (6) and (5),

$$B\big(\tilde\varphi(\varphi(z))\big) = v(\varphi(z))^{p+1} = v(z)^{p(p+1)} = B\big(\tilde\varphi(z)\big)^p = B\big(\varphi(\tilde\varphi(z))\big).$$

As B is injective on $D(0,r)$, we get $\tilde\varphi \circ \varphi = \varphi \circ \tilde\varphi$, so $C_{\tilde\varphi}$ commutes with C_φ. Since B^{-1} has a simple zero at the origin, $\tilde\varphi$ vanishes there to order exactly $q = (p+1)p^k$. Because $p < p+1 < p^2$, we have $p^{k+1} < q < p^{k+2}$, so q is not a power of p. Lemma 4 gives $C_{\tilde\varphi} \notin \overline{\mathrm{alg}(C_\varphi)}^{\sigma}$. □

4.3. **Lens maps.** Let $0 < \alpha < 1$, and let u, S and φ_α be as in the introduction. The inverse of u is $w \mapsto \tanh(w/2)$, and $\varphi_\alpha = \tanh(\alpha u/2)$. Put $T = C_{\varphi_\alpha}$. Since u is odd, so is φ_α, and hence R commutes with T.

Let us explain why R should not be in $\overline{\mathrm{alg}(T)}^{\sigma}$. In the coordinate $w = u(z)$ on the strip S, the operator T^k is composition with the dilation $w \mapsto \alpha^k w$, and R is composition with $w \mapsto -w$. As indicated in [2, Remark 8.12] and in condition (SK5) of [1], the proof of the theorem of Lacruz, León-Saavedra, Petrovic and Rodríguez-Piazza uses the composition operators induced by $u^{-1} \circ (\zeta u)$ for all ζ in an open set of parameters with $\zeta u(\mathbb{D}) \subset u(\mathbb{D})$, and analytic continuation in ζ; in [2, Theorem 7.1] translations play the role of dilations. For the strip, $\zeta S \subset S$ holds only for $\zeta \in [-1,1]$, so there is no room to continue from the positive dilations α^k to -1. We turn this into a proof by testing on a function h for which $T^k h$ and Rh, restricted to the radius $[0,1)$, are Cauchy kernels $1/(x - z_k)$ in the variable $x = u^3 \in (0,\infty)$, with poles $z_k = 64i\alpha^{-3k}$ for T^k and $-64i$ for R. It then suffices to find a function ω on $(0,\infty)$ whose Cauchy transform vanishes at the geometric sequence z_k but not at $-64i$. A function of $\log z$ with period $3\log(1/\alpha)$ and the right phase vanishes on the whole sequence z_k, and the factor $\exp(-2(-z)^{1/3})$ makes it decay fast enough to be a Cauchy transform of a function that defines a bounded functional on $H^2(\mathbb{D})$.

We will separate R from $\overline{\mathrm{alg}(T)}^{\sigma}$ by a functional $A \mapsto \langle Ah, g\rangle$, and the next lemma supplies the function on the half-line from which g is built. Put $\kappa = \pi/\big(3\log(1/\alpha)\big)$.

Lemma 5. *There is a function ω on $(0,\infty)$ with $|\omega(x)| \le M_\alpha e^{-x^{1/3}}$ such that the Cauchy transform*

$$G(z) = \int_0^\infty \frac{\omega(x)}{x - z}\,\mathrm{d}x, \qquad z \in \mathbb{C} \setminus [0,\infty),$$

satisfies $G(64i\alpha^{-3k}) = 0$ for every $k \ge 0$ and $G(-64i) = ie^{-8e^{i\pi/6}}\sinh(\kappa\pi) \ne 0$.

Proof. For $z \in \mathbb{C}\setminus[0,\infty)$ let $\mathrm{Log}(-z)$ be the principal branch of the logarithm, whose imaginary part lies in $(-\pi,\pi)$, and let $(-z)^{1/3} = \exp\big(\mathrm{Log}(-z)/3\big)$. Define

$$G(z) = \exp\big(-2(-z)^{1/3}\big)\sin\Big(\kappa\Big(\mathrm{Log}(-z) - \log 64 + \frac{\pi i}{2}\Big)\Big).$$

The argument of $(-z)^{1/3}$ lies in $(-\pi/3, \pi/3)$, so $\operatorname{Re}(-z)^{1/3} \geq |z|^{1/3}/2$. The imaginary part of the argument of the sine is at most $3\kappa\pi/2$ in modulus. Hence

$$|G(z)| \leq \cosh(3\kappa\pi/2) e^{-|z|^{1/3}}. \tag{7}$$

The boundary values $G_\pm(x)$ of G from above and below the positive half-line exist for every $x > 0$. Put $\omega = (G_+ - G_-)/(2\pi i)$; the bound on ω follows from (7). Cauchy's integral formula on the slit annulus $\{\eta < |\zeta| < \Xi\} \setminus [0, \infty)$, $\eta < |z| < \Xi$, gives $G(z)$ as the sum of the integrals over the two circles and of $\int_\eta^\Xi \omega(x)/(x - z)\,\mathrm{d}x$. As $\Xi \to \infty$ the integral over the outer circle tends to 0 by (7). As $\eta \to 0$ the integral over the inner circle tends to 0, since G is bounded near the origin: the first factor has modulus at most 1, and the argument of the sine has bounded imaginary part. Hence G is the Cauchy transform of ω.

If $z = 64i\alpha^{-3k}$, then $\operatorname{Log}(-z) = \log 64 + 3k \log(1/\alpha) - \pi i/2$, so the sine equals $\sin(k\pi) = 0$. If $z = -64i$, then $\operatorname{Log}(-z) = \log 64 + \pi i/2$, so the sine equals $\sin(i\kappa\pi) = i \sinh(\kappa\pi)$, while $(-z)^{1/3} = 4e^{i\pi/6}$. □

Proof of Theorem 6. Let

$$h(z) = \frac{1}{u(z)^3 - 64i}.$$

The roots of $w^3 = 64i$ are $4e^{i\pi/6}$, $4e^{5i\pi/6}$ and $-4i$, whose imaginary parts have moduli 2, 2 and 4. All three lie outside the closure of S, so $1/(w^3 - 64i)$ is bounded on S, and h is in $H^\infty(\mathbb{D}) \subset H^2(\mathbb{D})$.

Let ω be as in Lemma 5, and define a linear functional on $H^2(\mathbb{D})$ by

$$\ell(f) = \int_0^\infty \omega(x) f\big(\tanh(x^{1/3}/2)\big)\,\mathrm{d}x.$$

The norm of point evaluation at $r \in [0, 1)$ on $H^2(\mathbb{D})$ is $(1 - r^2)^{-1/2}$, and $1 - \tanh^2(y/2) = \cosh^{-2}(y/2)$, so

$$\big|f\big(\tanh(x^{1/3}/2)\big)\big| \leq \|f\|_{H^2(\mathbb{D})} \cosh(x^{1/3}/2).$$

Since $e^{-x^{1/3}} \cosh(x^{1/3}/2) \leq e^{-x^{1/3}/2}$ is integrable on $(0, \infty)$, the functional ℓ is bounded. By the Riesz representation theorem there is $g \in H^2(\mathbb{D})$ with $\ell(f) = \langle f, g\rangle$, and $\Lambda(A) = \langle Ah, g\rangle$ is continuous in the weak operator topology.

Let $r = \tanh(x^{1/3}/2)$, so that $u(r) = x^{1/3}$. The identity $u \circ \varphi_\alpha = \alpha u$ gives $u(\varphi_\alpha^{[k]}(r)) = \alpha^k x^{1/3}$, and hence

$$(T^k h)(r) = \frac{1}{\alpha^{3k} x - 64i} = \frac{\alpha^{-3k}}{x - 64i\alpha^{-3k}}, \qquad k \geq 0.$$

By Lemma 5, $\Lambda(T^k) = \alpha^{-3k} G(64i\alpha^{-3k}) = 0$ for every $k \geq 0$. On the other hand $u(-r) = -x^{1/3}$, so

$$\Lambda(R) = \int_0^\infty \frac{\omega(x)}{-x - 64i}\,\mathrm{d}x = -G(-64i) \neq 0.$$

Hence $R \notin \overline{\mathrm{alg}(T)}^{\sigma}$, as explained in Section 2. □

5. Concluding remarks

Vertical translations on $\mathcal{H}^2$. On $\mathcal{H}^2$ the operator C_φ with $\varphi(s) = s + i\tau$ is the unitary diagonal operator $n^{-s} \mapsto n^{-i\tau} n^{-s}$, so its spectrum is the closure of $\{m^{-i\tau} : m \in \mathbb{N}\}$, which is $\mathbb{T}$ for $\tau \neq 0$. The content of Theorem 1 is that the same holds in the Fréchet space $\mathcal{H}^\infty_+$, where the spectrum is not a priori closed. Since the function g in Theorem 1 is bounded in $\mathbb{C}_0$, the range of $C_\varphi - \lambda I$ does not even contain the space $\mathcal{H}^\infty$ of bounded Dirichlet series.

Symbols which are not univalent. The hypothesis of Theorem 2 is not empty. Let $\varphi(s) = s + c + a2^{-s}$, where $c > |a|$ and $|a| \log 2 > 1$. Then $\mathrm{Re}(c + a2^{-s}) > 0$ in $\mathbb{C}_0$, so $\varphi \in \mathcal{G}_{>0}$, while $\varphi'(s) = 1 - a(\log 2)2^{-s}$ vanishes at a point of $\mathbb{C}_0$. Theorem 2 does not address the following question, which concerns univalent symbols.

Problem 1 (Bayart–Yao [2, Question 4.10])**.** *Let $\varphi \in \mathcal{G}_{>0}$ be univalent. Under which conditions does C_φ have dense range?*

Diagonal operators and Sarason's theorem. The criterion of Theorem 3 is the diagonal case of Sarason's description of the weakly closed algebra generated by a normal operator [14, 15]: with $\mu = \sum_n 2^{-n} \delta_{\lambda_n}$ it says that the polynomials are weak-star dense in $L^\infty(\mu) = \ell^\infty$. The measure in the proof of Theorem 4 is Wolff's classical example [16]: in terms of the Cauchy transform, $\sum_j r_j^2/(z - c_j) = 1/z$ for $|z| > 1$, although the points c_j fill the disc. The existence part of [2, Question 5.6] can therefore also be read off from these classical results. What the proof above adds is the explicit form of the criterion for diagonal operators and a self-contained argument.

Local and global solutions of Böttcher's equation. The proof of Theorem 5 uses only that $\varphi(0) = 0$, that φ vanishes at the origin to some order $p \geq 2$, and that some iterate of φ has supremum norm less than 1. When ψ is zero-free, one may take $k = 0$ and replace v by the global solution of [2, Lemma 8.4]; this is the route of [2, Theorem 8.2].

Lens maps and unbounded Königs domains. Bayart and Yao [2, Remark 8.12] observed that in the theorem of Lacruz, León-Saavedra, Petrovic and Rodríguez-Piazza the boundedness of $u(\mathbb{D})$ can be replaced by two conditions: that a certain set Ω of parameters is open, and that $\|u^n\|_{H^2(\mathbb{D})} \leq C^n$ for every $n \geq 1$. They also noted that the second condition fails for the lens maps. Indeed, such a bound would give $|u(z)| \leq C$ for every $z \in \mathbb{D}$ after taking nth roots in the point evaluation estimate. The set Ω is printed as $\{w \in \mathbb{C} : w + u(\mathbb{D}) \subset \mathbb{D}\}$; for the strip, both natural readings give a set with empty interior, since $\{w \in \mathbb{C} : wS \subset S\} = [-1, 1]$ and $\{w \in \mathbb{C} : w + S \subset S\} = \mathbb{R}$.

In a later paper, Bayart, Wang and Yao [1, Theorem 6.3] recover the theorem of Lacruz, León-Saavedra, Petrovic and Rodríguez-Piazza, and

they remark [1, Remark 6.4] that the boundedness of the range of the Königs function can be replaced by the assumption that every power u^n belongs to $H^2(\mathbb{D})$, which holds whenever C_φ is compact. The lens maps are univalent and fix the origin, their Königs domain S is strictly starlike with respect to the origin, u^n is in $H^2(\mathbb{D})$ for every $n \geq 0$, since u has only logarithmic singularities at ± 1, and C_{φ_α} is compact, indeed in every Schatten class [10, Theorem 2.1]. Hence Theorem 6 shows that the boundedness of $u(\mathbb{D})$ cannot be replaced by this assumption alone. The theorem with the bounded range hypothesis is not affected. In the framework of [1], conditions (SK3) and (SK5) fail for the lens maps, by the observations of the preceding paragraph.

Finally, for $\alpha = 1$ the lens map is the identity, and $C_{\varphi_1} = I$ does not have a minimal commutant either, since $\{I\}'$ contains every bounded operator while $\overline{\mathrm{alg}(I)}^{\sigma} = \mathbb{C}I$.

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